

Commentary

On Being a “Critic and Conscience of Society”: The Role of the Education Academic in Public Debates

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There is a long history of the “public intellectual”. Emerson (1837) defined the “American Scholar” as a person enriched by the past, who communicates ideas, and who is the “world’s eye.” In his opinion, the “most important activity is action – inaction is cowardice.” More recently, Said (1993) argued that academics need to stand outside society and its institutions and actively disturb the status quo but at the same time address their concerns to as wide a public as possible. They must, Said argued, resist being co-opted by those in power and stand up for the universals of truth and justice at all costs.

In New Zealand, academics are empowered by legislation to be the “critics and conscience” of society. Today, this is carried out mainly in fora such as academic journals, where the intended audience is other academics and students rather than power brokers, policy makers, and the taxpaying public. Of course academic journals are an important forum; they allow “[c]ritical ideas and alternatives ... [to] be advanced according to the scholarly canons of respectful, evidence-based exchanges” (Fallis, 2004, pp. 47-48). However, while academics can, and must, use the peer review world of journals and books, the effect of academics’ ideas is often lessened because of the relative inaccessibility and lack of timeliness: too often academic policy issues seem more historical accounts of what has been than what could be. It is my belief that it is important for academics to speak out on topics within their expertise and take seriously their role as critics and conscience of society, and that this involves not only writing in academic journals but also making the academic voice visible to a wider, more general audience by using the media.

But media coverage can mean issues are stripped of complexities and nuances, and controversies presented as black and white. The media requires immediate reaction to issues, which can be at the expense of informed response. And while new media technologies provide more opportunity for public debate and critique than the average educational journal, they lack the checks and balances that come from academic peer review. Nor is it uncommon in both conventional and new media for the person, rather than the ideas to be subjected to scrutiny, misrepresentation, even personal attacks. Thus, the dangers inherent in using the media, such as the risk of trivialisation or sensationalisation of issues, is compounded by the ease which critics can uncritically respond through new media such as blogs, where there can be limited means of redress, especially when the attacks are more about the person than the ideas.

This raises the question, then, of how best academics can be involved in public critique in a timely and informed manner, while minimising the personal attacks and misrepresentation (see also Simmons, 2007). This paper uses my experiences over the past few years to discuss notions of critique in public debates, and the tensions and responsibilities when becoming involved in making statements about public policy.

The experiences of one public academic

Over the years, I have been prepared to speak about my area of expertise, although I have clear boundaries about what and where I am willing to comment: only speak about that in which I have expertise; base it on my reading and interpretation of the research; and aim always to enhance my profession by demonstrating respect and caution, while being mindful of the possible intended and unintended consequences of commentary. I have had media exposure before the publication of my book *Visible Learning: A Synthesis of over 800 Meta-analyses Relating to Achievement* (Hattie, 2009d). But it was the January 2009 Sunday Star Times newspaper front page article about the release of *Visible Learning* that increased the exposure, followed by extensive media coverage of my later comments about national standards.

I had been working on the ideas in *Visible Learning* for the past 15 or more years. It involved synthesising more than 800 meta-analyses of research on the influences on achievement for school-aged students. For all 138 topics in the book, I had read much of the primary literature referenced in the meta-analyses, and, indeed, in some areas, had conducted studies myself. I became particularly fascinated with why most of our debates about schooling concerned the things which the evidence shows have the least effect, particularly structural matters such as grouping or class size. This fascination led to more in-depth studies and separate publications on streaming (Hattie, 2002), and class size (Hattie, 2007). The synthesis of meta-analyses showed the enormous power of quality teaching, that variance among teachers is one of the most critical influences on student learning and achievement, and that there are strategies that seem to invoke more learning than others. But these practices are harder to legislate for and to change, and thus are too often overshadowed by changes to class sizes or streaming, which are easier to effect.

I asked in *Visible Learning* why we keep defending the rights of teachers to choose their own teaching methods, why we continue to allow the “everything goes” notions of teaching, and why there are never any “below average” teachers. The answer given by the synthesis of meta-analyses was that it is primarily because we set our standards so low – it seems it is sufficient to ask for a level of evidence so low that any teacher or school could show they have enhanced learning. In the metric of meta-analyses, this is akin to saying that if you can get an effect greater than zero (i.e., enhance learning by any amount, $d > .0$), then this is sufficient. If the measure of change is set as “anything above 0 effect size”, then 97%-plus of teachers can meet this grade and obviously everything these teachers do is supported by evidence (when the bar is so low) and all are above average (if “average” is set at any degree of change). The problem is that the average effect of all influences (at least based on 800 meta-analyses and about 200 million students) is much greater than $d = 0$. There is a normal distribution of effects centred on an effect size of about $d = .40$. Thus, teachers who only attain up to $.40$ are those “below average” in their influence, and

there are just as many teachers who attain effects greater than .40 as there are those below .40. I labelled this .40 as the "hinge-point" and then set out to understand the common features of the influences above compared to those below this hinge-point.

I had tried out my story (about the power of feedback, challenge, evidence of impact on students, etc.) in many seminars, conferences, and workshops both in New Zealand and overseas, and a workbook for teachers (Clark, Timperley, & Hattie, 2003) had already introduced many of the ideas in *Visible Learning* (e.g., the power of learning intentions, success criteria, feedback, and just-in-time assessment to inform teachers of their effects). So when the "story" contained in *Visible Learning* was headlined in the Sunday Star Times in January 2009, it would not have been news for many in education in New Zealand. Nonetheless, the reaction was intense.

It was clearly evident that many of those commenting had not read the book (some even applauded that they had not read it). There were outcries that class sizes must be reduced and that it was obvious that I must be wrong (even though I never claimed that reducing class size did not lead to enhanced achievement); that performance pay was evil; that streaming was easier for teachers as there were fewer diverse students in a class; and that there were instances where retaining a student was beneficial. Anecdotes and war stories abounded, as if each were counter-factual. The story became that of "topic headlines" and the league table of influences, and my own carefully crafted story based on interpretations of the extensive data base in the book dipped below the horizon. The rage about the book continued for many months.

Alexander (2010), who has released various reports and books on the status of education in England, has noted changes in the nature of critique. He claims that reactions to the reports have moved: "In place of a discourse likely to create policies which have the best chance of success with the minimum of collateral damage, we have four corrosive pretenders, the discourses of dichotomy, derision, myth and meaninglessness." The discourse of dichotomy reduces everything to mutual exclusives, the discourse of derision defines the tone of response ("first misrepresent then ridicule it, personalizing the attack where possible and appealing to the lowest common denominator of popular prejudice"). The discourse of myth played out by many critics is that "it was not like this in my day", "I did not teach this way", "we did not have jargon or management-speak in our day", and a general amnesia to the difficulties of implementing any positive change. The discourse of meaninglessness is about the destruction of language itself - "new initiatives, step changes, hitting the ground running, driving up standards, rolling out innovation, zero tolerance, best practice, world class schools, back to basics and the rest".

Critique of ideas is not only a form of feedback but also the essence of academia, and can lead to debate that is more informed. The fundamentals of academic critique are that it relates to the idea not the person; takes careful consideration of the tenets and assumptions; and is aimed at informing and extending the debate - not shutting it down because the ideas are not welcome or are not part of some critics' dogma. Critique should not raise false premises, or champion the detail over the message, and typically it requires at least a modicum of courtesy and goodwill to further the debate.

The next sections outline various types of critique. Reactions to *Visible Learning*, as well as to my public comments on National Standards, are used to illustrate these types and to clarify some of the misconceptions.

Critique of research method

I comment in the foreword to *Visible Learning* that the book “is not a book about criticism of research and I have deliberately not included much about moderators of research findings based on research attributes (quality of study, nature of design), not because these are not important but because they have been dealt with elsewhere by others (e.g., Lipsey & Wilson, 1993; Sipe & Curlette, 1996a, 1996b)”. But this comment seems to have been misinterpreted as my paying no attention to the quality of the research (Snook, O’Neill, Clark, O’Neill & Openshaw, 2009, for example). Where research quality relating to specific influences has been shown to make a difference, this has been cited. Where a meta-analysis itself was poor quality, I was not backward in saying so. It would be hard, for example, to ignore the dreadful state of the research on learning styles, instances of strange findings that did not fit with the majority of studies in a domain, or instances where moderators were so powerful that the average was not convincing (e.g., homework). Thus, claims that the studies were not appraised for their validity are misleading and incorrect. One of the very powers of meta-analysis is to deal with this issue. Readers and policy makers can have assurance that the conclusions I made are based on “studies, the merits of which have been investigated”. I did not write a book about research design.

Another major set of criticism is about the nature of the research used in the book. Meta-analyses are a form of synthesising past literature in a systematic manner and by its nature exclude qualitative research. Meta-analyses involve calculating effect sizes (standard deviation units) based on data from the original articles and then asking about possible influences that may moderate these effects. The literature on the topic is large and it did not seem necessary to repeat the well-rehearsed arguments for and against meta-analysis in detail. However, the second chapter of the book did outline the limitations as they pertained to my study and ironically, these now seem to be repeated (sometimes verbatim) by critics as if I had not realized them.

Critique of bias and “it does not apply here”

Snook et al. (2009) claimed that I did not acknowledge that “bias is not normally controlled in meta-analysis”, but they failed to acknowledge the rich literature on this topic which shows how controlling bias is a major concern of this method (Schulze, 2004). Indeed, as Snook et al. themselves noted, there is a heavy reliance on published studies – which, through the use of peer reviewing, is one way of maximising quality control. To then claim that I ignored the many potential influences (such as quality, year of schooling) on the overall conclusions is not correct, given I was careful to note these effects whenever they occurred and were relevant to the pros and cons of the arguments developed throughout the book. The very case they cite, homework, is one such case where it was noted that there was an important moderator (primary vs. secondary schooling). There have been media

reports claiming that I argued that homework does not make a difference to achievement, but to confuse such second hand accounts as my own views is misleading. My argument is that the past review of homework, especially in primary school, has shown that many of the methods used in homework have not had, on average, much of an effect relative to other effects; and I have encouraged schools to consider alternative methods to ensure more effective use of homework in learning (and advised schools not to abandon homework, because so many parents judge the quality of teachers and schools by the presence of homework).

A further criticism has been about the generalisability of the studies – with the suggestion that overseas research could not possibly apply to New Zealand. I noted in the book that most of the studies were carried out in highly developed English-speaking countries (mainly the United States and the United Kingdom) and should not be generalised to non-English speaking or developing countries. It has been shown, for example, that in developing countries, school effects (as against teacher effects) are huge, because of the wide variety and quality of schools. Snook et al. claimed that New Zealand may be so different from the rest of the Western world that "we should exercise great care in relating the meta-analyses to New Zealand education." It could be that New Zealand schools, teachers and students are indeed different from those of the United States. However, the criticism should have been that if there is not sufficient New Zealand research to ascertain the differences or the effect of various influences, then there may be merit translating the existing research rather than to rely on no research at all! That the studies are not based on rural New Zealand small town Year 5 students, for example, does not mean teachers of such classes should dismiss the messages in the book as irrelevant to them.

There is also the criticism by Snook et al. that I adopted an "arbitrary" cut off of .40 effect-size – indeed, perhaps it was arbitrary, but it was certainly not capricious. This cut was the average effect from 800-plus meta-analyses and the book fully elaborates on the origin of this claim. It is a threshold to create a story (about what is common above and below this effect) but it is not presented as a fixed cut-score below which one should ignore and above one should cherish – many parts of the book go to some length to show the importance of not making such conclusions. There is also misinterpretation of the .17 effect from Outward Bound programs in which critics have argued that I ignored my own .40 "hinge point" – the exciting aspect of this effect is that it is a rare case of an additive effect over time: the .17 effect I was excited about was additional to the program's immediate .60 effect on academic achievement (that is, the effect, months later, rises to .77).

Critique of exclusion

Among the most common criticism of *Visible Learning* is the place of non-school factors. I indicate early on that such factors, while important, are not the major focus of the book (although there is a chapter on "what the child brings to the classroom" from the home, social background, and so on). I realised that many sociologists would not be pleased that non-school factors were not placed at the forefront of all claims, but I noted that this was "NOT because they are unimportant, indeed they may be more important than many of the issues discussed in this book. It is just that I have not included these topics in my orbit." (pp. x-ix). As educators, we have little

control over these factors and I constrained my interest primarily to what we can have control over. There seems no doubt that changing the conditions of the home (poverty, parental expectations, and so on) could lead to major differences to students, and indeed there are sections in the book on these issues.

Critique of narrowness in outcome

There has been concern about the emphasis in *Visible Learning* on achievement. As I noted in the book, it is but one of the many outcomes of schooling, but surely it ranks high in desired effects of schooling. Snook et al. go further and claim that I am more concerned with surface level achievement. Ironically, this is my criticism of the research. Not only is much research based on surface achievement – so are most teacher-made tests and teacher classroom questions. I note that surface knowledge is not “bad” but that we should be more attentive to the proportions of surface, deep, and conceptual knowledge.

Critique in support of favoured topics

The most contentious counter claims to *Visible Learning* have related to class size. Minto (2009) argued that claims about class size are “extremely suspicious because there have never been properly controlled studies anywhere in the world where similar groups of students have been taught the same material by similar teachers in classes of 15 students compared to 30 for example.” While this did not stop him making claims that reducing class size would make a difference, he seems to overlook the Tennessee class size study, which is one of the most well known randomised control studies (of course, with criticisms). Snook et al. were equally selective and cite one study where the effects were large (.62) but ignore that, with an average effect of .20, there must also have been large negative effects to get to the average of about .20. The major advantage of meta-analysis is that extreme findings are moderated by a larger sample of studies, even though we may learn much from the extreme studies.

What I said in *Visible Learning*, in fact, was that the average effect of class size ($d = .20$) shows that reducing class size (usually from 25–30 to 15–20) does enhance achievement, but relative to all possible effects (average $d = .40$), it was possibly not worth the millions of dollars it would take to reduce class sizes, and which could well be spent on other more powerful effects. The criticisms by Snook et al. are details well rehearsed in the literature on class size (including in *Visible Learning* and Hattie, 2007). It is indeed the case that small and large are not specifically defined as one set of numbers (in Hong Kong, for example, they are aiming to reduce classes from more than 50 to classes of about 30 students); that teacher/student ratio is not the same as class size (noted in almost every study on this topic); that attendance in class is important (it is even for those effects well above .40!), and so on. Meta-analysis indeed does not ignore such problems and one of the assets of this method is that these influences can be evaluated for their effects.

Of course, the aim is to find the optimal conditions for ensuring that reducing class size can make a difference. Snook et al. claimed that well designed studies have higher effects, that it may be higher in some subjects than others, and so on. They cite research in the United Kingdom by Blatchford (2003), who said previous studies

were of low research design, and that he found many positive effects for small class size in the early years of schooling. But these studies and moderators (e.g., year of school, the quality of study) were included in my research syntheses – the average effect still was .20 and there were no effects moderated by quality or year of school. It is not correct to see meta-analysis as a blanchmange where all is combined and details are not explored. Furthermore, I looked at the reasons class size reduction did not lead to greater effects and the explanation was that teachers too rarely varied their teaching to take advantage of smaller class sizes. Hence the conclusion, that if a policy was to reduce class size, it should be accompanied by different teaching (a view also endorsed by Snook et al.).

Critique based on pet hates

Many critics had a pet peeve about any discussion or support of research on performance pay. My own work with the National Board for Professional Teaching Standards, the United States National Council for Accreditation of Teacher Education, and other groups investigating these effects has demonstrated the difficulties of relating teacher income to performance and indeed I have published on these difficulties (Hattie & Clinton, 2008; Invgarson & Hattie, 2009). I still would argue that it is appropriate and worthwhile to question how to differentially pay teachers particularly based on some component of expertise. This is a well-rehearsed practice in many disciplines and vocations. The usual argument against it in education – that it would lead to competition in staff rooms if teachers were differentially paid – fails to recognise that they are already paid differentially (mostly on experience) in any one staffroom. While the perfect solution is not yet before us, this should encourage us to keep questioning and researching ways to address this issue.

Critique of the role of advisor

Being an advisor on policy is a well-researched domain (Scott, 2005) although seemingly less known in education. The Westminster system of government means advisors are apolitical and serve successive governments irrespective of political pedigree. Governments can set policy direction as the National-led coalition government has with respect to National Standards. The task then for agencies is to either manage or minimize risk, and for an independent advisor, it is to provide quality advice to help decision makers choose policies and associated interventions that support strategic directions and minimise damage while leading to the desired policy outcomes.

Over my decade in New Zealand, I have been asked by many politicians from all sides to meet with them about education matters. The crucial aspect is to speak to them about research, not to uncritically advocate a policy. I met with the then shadow Minister of Education (Bill English) about testing and standards, and with the then Leader of Opposition (John Key) and other ministers of both parties about many educational matters. I have sent them articles, shown them the work we are doing, and suggested they visit schools in New Zealand that exemplify what I consider to be the very best of implementation of excellence. I have long worked in the research domain of goals, target setting, adding value, transparent learning

intentions and success criteria, and expectations and thus see many merits of a policy based on standards. The issue relating to national standards that has most concerned me is implementation and that has been my constant message.

I have had little to no involvement in the implementation of national standards over the past year. I was invited to be part of a technical advisory group which met on two occasions. But given I was out of the country for six months, I attended neither and there have been no minutes or papers from those meetings. So zero influence there. A few years ago, five friends with expertise in measurement set up a group, the New Zealand Assessment Academy, to discuss research on measurement issues. While there is much assessment debate in New Zealand, there is far less interest in measurement (the more technical aspects of assessment). The Academy was invited to comment on some aspects of the standards – and we indeed provided critiques to the Ministry of Education about the standards, standard setting, and other assessment matters. These were part of many other inputs the Ministry received, and some, but by no means all, of our advice has been enacted.

When the Prime Minister, John Key, launched the standards he did acknowledge my involvement in promoting the notion of standards and how assessment can influence the nature and quality of teaching and learning. This seemed to many people to indicate that my hands were “on the tiller”. Yes, I did advise that national standards that build on the current curriculum can help redress a major problem in New Zealand schooling – the problem of ensuring that teachers had a common conception of progress. After the launch, I set my views in print, as would be normal in academia. These were placed on the Cognition web site (www.cognition.co.nz), at that time being launched as a place for policy debate. These comments were picked up by the media and the message was that I was now a critic; many wanted a black or white answer – was I for or against standards – and thus they failed to see my direction. But providing black and white answers is not the nature of academic critique. I remain concerned with their implementation and my constant message is that any implementation must gain the confidence of, and share the ownership with, those who have to deliver the policy. It seems a truism that a policy has a higher success when it wins the hearts and minds of those delivering the policy. Under Tomorrow’s Schools there is much discretion given to schools as to how to implement, and this seems to be ignored in the recent pronouncements about compliance to the policy (see Hattie, 2009c).

Despite our high rankings on international league comparisons, there are some problems in New Zealand which we know well and need to address immediately. National Standards can provide useful evidence and can influence issues such as: ensuring there is greater emphasis in schools (and in reports to parents) on progress in, as well as levels of, achievement; ensuring that teachers have a common conception of progress (e.g., teachers trusting the information coming from previous teachers both within and across schools); enhancing the already developing consistency of interpretation of levels within the New Zealand curriculum; demonstrating that all students are making appropriate progress (as this is the optimal way to reduce the disparities – the aim should NOT be to reduce the 20% tail, because if we are successful, there would still be 20% in the tail; see Hattie, 2009b); sharing evidence-based practices within and across schools; enhancing

student assessment capabilities about their levels of performance and progress; and identifying schools where students may not be attaining satisfactory progress and intervening more quickly than we do now. Most important, these factors can lead to more effective teaching of all students, as the only way the policy is going to be effective is if the teaching is more appropriate, rigorous and shown to be effective in assisting students reaching the targets (both the levels and the progress) for all students.

The dilemma I am in, given I am an academic researching in these areas of assessment, teaching and learning, is how to be part of a solution while remaining a critic and conscience of society. A few years ago, I was asked to be part of a Ministerial advisory panel for scholarship, and the aim was to advise the Minister (and others) of ways to improve the assessments. This we (Terry Crooks, Gary Hawke and myself) did, and then our brief was extended to NCEA (and the group enlarged). Similarly, I have joined a similar group regarding National Standards – with the aim of giving independent advice to the Minister on the technical and educational aspects of the standards that may lead to refinements or changes to be made to the design of the standards. We have set up various ways to seek independent views from the sector, and have a work program to be fully aware of implementation issues in standards debates. This places me within the “system” but as a Ministerial advisory group we are outside the Ministry. This can be a tough line to walk. I only hope that my transparency of views (published in various places), continued writings and speaking will continue to demonstrate my view, and that the involvement and actions in the advisory group demonstrates that I am prepared to call the situation as my research experiences tells me. Being part of the solution is a lot harder than critiquing from the side – and both views, in my opinion, are critical to a worthwhile society.

Critique of black and white

The critique that has come from this involvement as advisor is particularly vehement from those who do not agree with standards and where any involvement to make them work is anathema to their views. It has led to uninformed comments, personal attacks, and distrust from those who have not read my views or heard me speak. Fervent people often like their ideas as black and white. This is not, however, the way ideas work – they do not fall so simply one way or the other. That, for example, I can see many merits in the policy of national standards but have concerns about the implementation seems to baffle some critics (see Hattie, 2009a for an outline of my views on National Standards).

Critique of the personal

There are bloggers who prefer to cast personal aspersions. I have been accused of being a mad scientist, slippery as snake slither, and desiring to have “increased centralised control through testing” – whereas I have long argued publicly, to Parliament enquiries, and in my publications against this very notion. Such personal attacks, non-sequiturs, and extremism should not need counter arguments but they do illustrate the reactions to academics who make public comment. When they are

then cited by other bloggers, there is a false appeal to the scholarly tradition of citation.

Critique based on claims of corruption and conflict of interest

A theme in some of the criticism is that there is a money making purpose to my academic research. O'Neill (2009), for example claimed that the *Visible Learning* Laboratories, of which I am a director, is a commercial company (it is not), that I have made Ministry of Education policy (I have not), and that somehow I am making money out of the ventures (except for book royalties, this is incorrect). He has claimed in multiple sources (e.g., PPTA News, 2009) that I (and colleagues) wrote a "policy paper" on directions for assessment in New Zealand. This paper was commissioned by the Ministry of Education to suggest a direction that policy may address. It was made clear to us that we were not writing policy. It is hardly unusual in the Westminster system of governance for ministries to ask for academic views and input, but this does not make them policy. We were chosen because of our assessment expertise, but as the opening statement in the paper notes: "The purpose of this paper is to provide broad advice to the Ministry of Education to guide and inform the design of new and improved strategies, policies, and plans for assessment." Advice not policy.

Conclusions

If academics are to have a more immediate effect on public discourse and policy than what can be accomplished from writing in learned journals and other scholarly outlets, there is a price to pay in this world of immediacy of opinion, ready access to personal views, and the changing notions of critique. But to be out of the domain of influence – influence of exposing the many forms of evidence in an "evidence-based" era - is to reduce the purpose and power of academic research and teaching. There needs to be a renewed debate about the role of academics in public fora, in policy commentary, and how best they are to deserve their place among the "critics and conscience" of society. Of course this role is not the privilege of academics only, but the nature of the discourse, particularly in the non-quality-reviewed sphere of the blog-space, means that the debates are more personal, harder to argue against, and often flimsy and inaccurate (with little chance of correction). There is the expected added weight of criticism, some ill-informed and personal but mostly respectful evidence-based exchanges, and, as Gattone (2006) noted, "in spite of these pressures, social scientists continue to possess the ability to approach their work as public intellectuals. Their connections to empirical reality and their skills in the symbolic domain render them uniquely able to frame information in ways that help others make sense of the world" (p. 145).

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